Ema Lovrić, 10.9.2026.

Assignment 19: Windows system performance and resource analysis

https://github.com/elovric13/eutech/tree/windows-system-performance-and-resource-analysis-elovric13

1. Identify and record the following information about your laptop:

- Windows version.

OS Name	Microsoft Windows 11 Pro
Version	10.0.26200 Build 26200

- Processor.

Processor	Intel(R) Core(TM) i7-1065G7 CPU @ 1.30GHz, 1498 Mhz, 4 Core(s), 8 Logical Processor(s)

- Installed RAM.

Installed Physical Memory (RAM)	12.0 GB

- Storage capacity and available space.

Item	Value
Drive	C:
Description	Local Fixed Disk
Compressed	No
File System	NTFS
Size	475.75 GB (510,837,211,136 bytes)
Free Space	36.80 GB (39,518,294,016 bytes)
Volume Name	
Volume Serial Number	F810B2F7

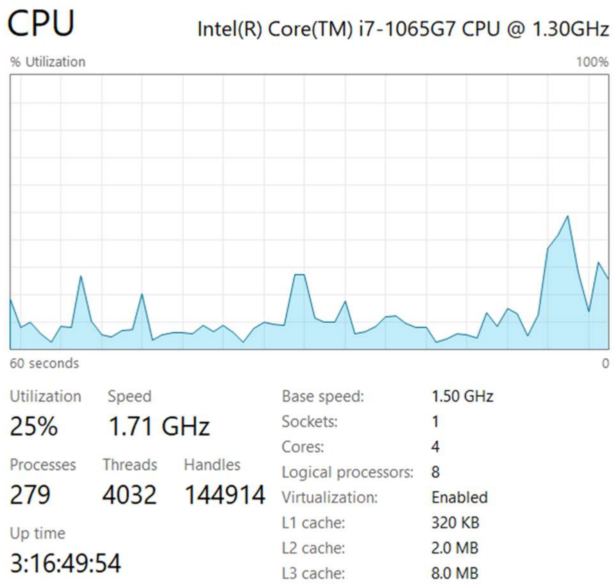
- System architecture.

System Type	x64-based PC

2. Open Windows Task Manager and analyze:

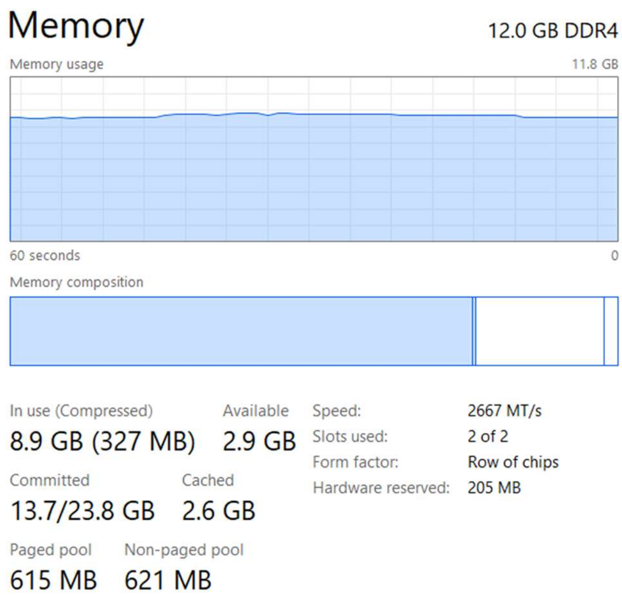
- **CPU usage.**

At the time of analysis CPU utilization was between 6% and 25% depending on the moment of capture, running at approximately 1.52–1.71 GHz out of a base speed of 1.50 GHz. 279 processes were running with 4032 threads and 144,914 handles. System uptime was 3 hours 16 minutes. The CPU graph shows irregular spikes, indicating short bursts of activity rather than sustained high load.



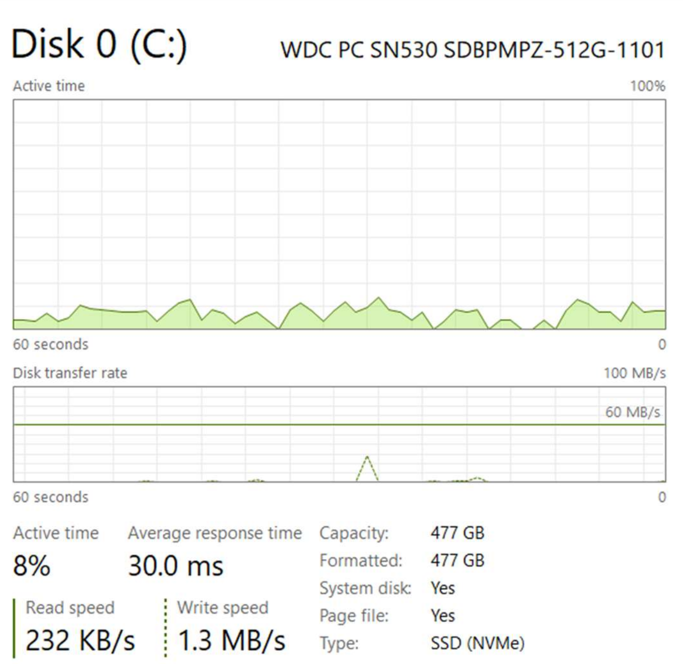
- **Memory usage.**

8.9 GB out of 11.8 GB usable RAM was in use at the time of analysis, representing 75–77% memory utilization. Available memory was 2.9 GB. Committed memory was 13.7 GB out of a maximum of 23.8 GB, meaning the system was using the page file to compensate for physical RAM pressure. RAM speed is 2667 MT/s DDR4, using 2 of 2 slots.



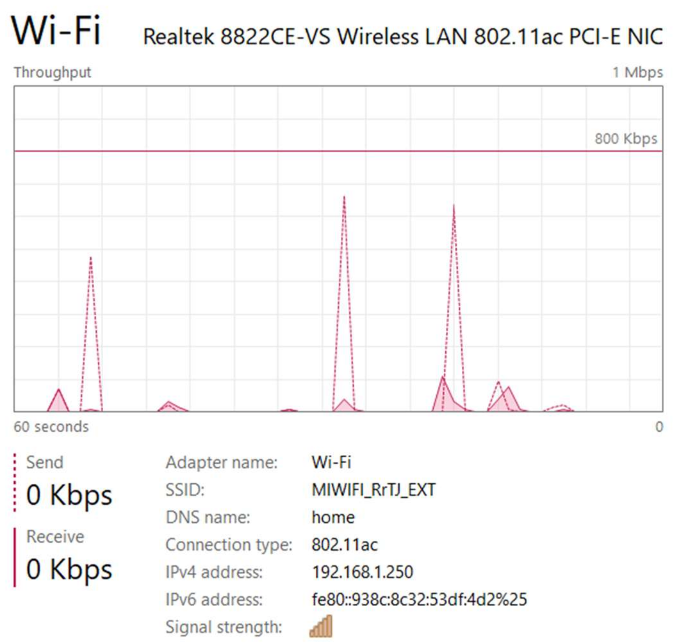
- **Disk usage.**

Disk 0 is a WDC PC SN530 NVMe SSD with 477 GB capacity. Active time was 8% with an average response time of 30.0 ms. Read speed was 232 KB/s and write speed 1.3 MB/s at the time of capture, indicating light disk activity.



- **Network activity.**

The active network adapter is a Realtek 8822CE-VS Wireless LAN 802.11ac connected via Wi-Fi. Send and receive were both 0 Kbps at the time of capture, with occasional spikes visible in the throughput graph. Four additional VMware virtual network adapters were present but showing no activity.



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3. **Identify the five processes currently consuming the most system resources and record their CPU and memory usage.**

Top five processes by resource usage – sorted **by CPU**

Process	CPU	Memory
Task Manager	1.3%	93.3 MB
Google Chrome (19)	0.6%	797.4 MB
WhatsApp (9)	0.4%	677.8 MB
Microsoft Edge (14)	0.4%	143.6 MB
Certilia Dcs (32 bit)	0.2%	0.9 MB

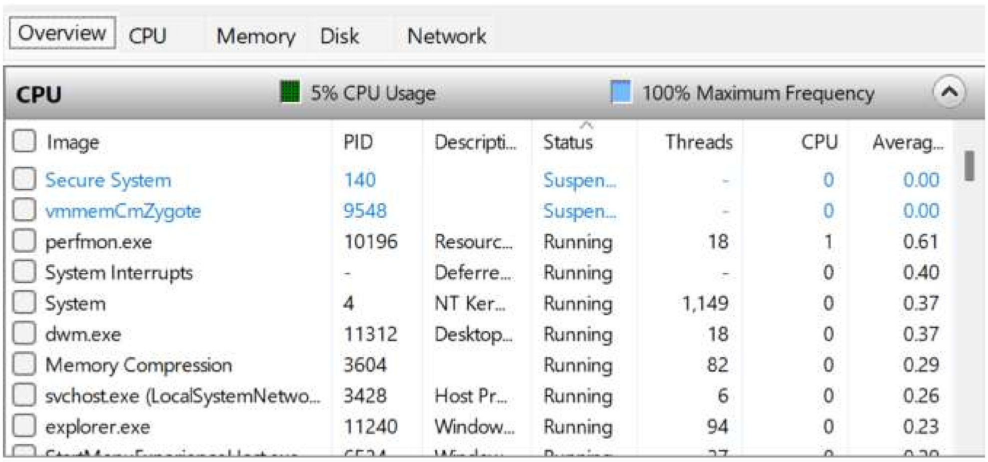
Top five processes by resource usage – sorted **by memory**

Process	Memory	CPU
Google Chrome (19)	797.4 MB	0%
WhatsApp (10)	724.7 MB	0%
Claude (9)	373.4 MB	2.1%
trsws_rest_api_1.4.0	255.4 MB	0%
Antimalware Service	174.2 MB	0.4%

4. **Open Resource Monitor and examine CPU, memory, disk, and network activity in greater detail.**

- **CPU**

Overall CPU usage was 5–14% during the observation period, running at 100% maximum frequency. The most active processes were perfmon.exe (Resource Monitor itself), System Interrupts, System, dwm.exe (Desktop Window Manager), and Memory Compression. No process showed critically high CPU usage, confirming the system was under light load during the analysis.



Overview CPU Memory Disk Network

CPU 5% CPU Usage 100% Maximum Frequency

Image	PID	Descripti...	Status	Threads	CPU	Averag...
Secure System	140		Suspen...	-	0	0.00
vmmemCmZygote	9548		Suspen...	-	0	0.00
perfmon.exe	10196	Resourc...	Running	18	1	0.61
System Interrupts	-	Deferre...	Running	-	0	0.40
System	4	NT Ker...	Running	1,149	0	0.37
dwm.exe	11312	Desktop...	Running	18	0	0.37
Memory Compression	3604		Running	82	0	0.29
svchost.exe (LocalSystemNetwo...	3428	Host Pr...	Running	6	0	0.26
explorer.exe	11240	Window...	Running	94	0	0.23

- **Disk**

Disk I/O ranged from 84–588 KB/sec with highest active time of 1–3%. The most active processes accessing the disk were msedge.exe, claude.exe, System, and svchost.exe. Response times were in the normal range. The NVMe SSD handled all disk operations without any bottleneck.

Disk	588 KB/sec Disk I/O			3% Highest Active Time			
Image	PID	File	Read (...	Write ...	Total (...	I/O Pri...	Respo...
msedge.exe	10084	C:\$L...	0	202	202	Normal	19
claude.exe	12340	C:\$L...	0	768	768	Normal	19
System	4	C:\Us...	0	6,144	6,144	Normal	12
System	4	C:\Wi...	0	77	77	Normal	12
System	4	C:\Us...	0	236	236	Normal	12
msedgewebview2.exe	29720	C:\Us...	1,311	0	1,311	Normal	11
svchost.exe (netsvcs -p)	20016	C:\Wi...	32,768	0	32,768	Normal	11
dwm.exe	11312	C:\Wi...	2,731	0	2,731	Normal	11
System	4	C:\Wi...	0	1,214	1,214	Normal	11

- **Network**

Network I/O was minimal at 0–2 Kbps with 0% network utilization overall. The most active network processes were MsMpEng.exe (Windows Defender, sending telemetry), claude.exe (active connection to Anthropic servers at 160.79.x.x), and svchost.exe (Windows services). LSBUpdater.exe was also visible with minor network activity.

Network	2 Kbps Network I/O			0% Network Utilization	
Image	PID	Address	Send (B/sec)	Receive (B/...	Total (B/sec)
MsMpEng.exe	12192	48.209.10...	9,327	2,695	12,023
claude.exe	12340	160.79.10...	6,552	3	6,555
svchost.exe (netsvcs -p)	3980	20.190.17...	1,355	4,160	5,515
LSBUpdater.exe	23504	a23-14-13...	78	673	751
msedge.exe	10084	150.171.2...	60	201	260
msedge.exe	10084	a96-16-86...	224	0	224
msedge.exe	10084	DESKTOP-...	0	181	181
UDClientService.exe	5412	server-13-...	35	117	152
claude.exe	12340	DESKTOP-...	0	126	126

- **Memory**

Physical memory usage was consistently at 58–59% during Resource Monitor observation. Hard faults ranged from 0–4 per second — occasional but not concerning. The highest memory consumers were MsMpEng.exe (445 MB committed), claude.exe instances (multiple processes totalling significant memory), msedge.exe, and explorer.exe. Memory Compression was active, compressing infrequently used data to free up RAM.

Memory	0 Hard Faults/sec			59% Used Physical Memory		
Image	PID	Hard Fa...	Commit...	Workin...	Shareab...	Private (...
Memory Compression	3604	0	3,760	278,444	0	278,444
MsMpEng.exe	12192	1	445,340	393,184	164,776	228,408
claude.exe	27396	1	236,016	318,228	106,092	212,136
claude.exe	19772	0	172,516	223,624	77,148	146,476
msedge.exe	15236	0	146,048	172,616	46,708	125,908
claude.exe	21704	0	185,832	219,472	104,556	114,916
explorer.exe	11240	1	157,724	345,232	230,516	114,716
TabTip.exe	17048	0	113,944	167,472	59,324	108,148
claude.exe	15188	0	116,340	222,388	117,060	105,328

5. **Open several applications that you normally use on your computer and observe how resource usage changes. Compare CPU and memory usage before and after opening these applications.**

To observe the effect of opening additional applications, resource usage was recorded before and after launching Spotify, Google Chrome, and Microsoft Edge with additional tabs.

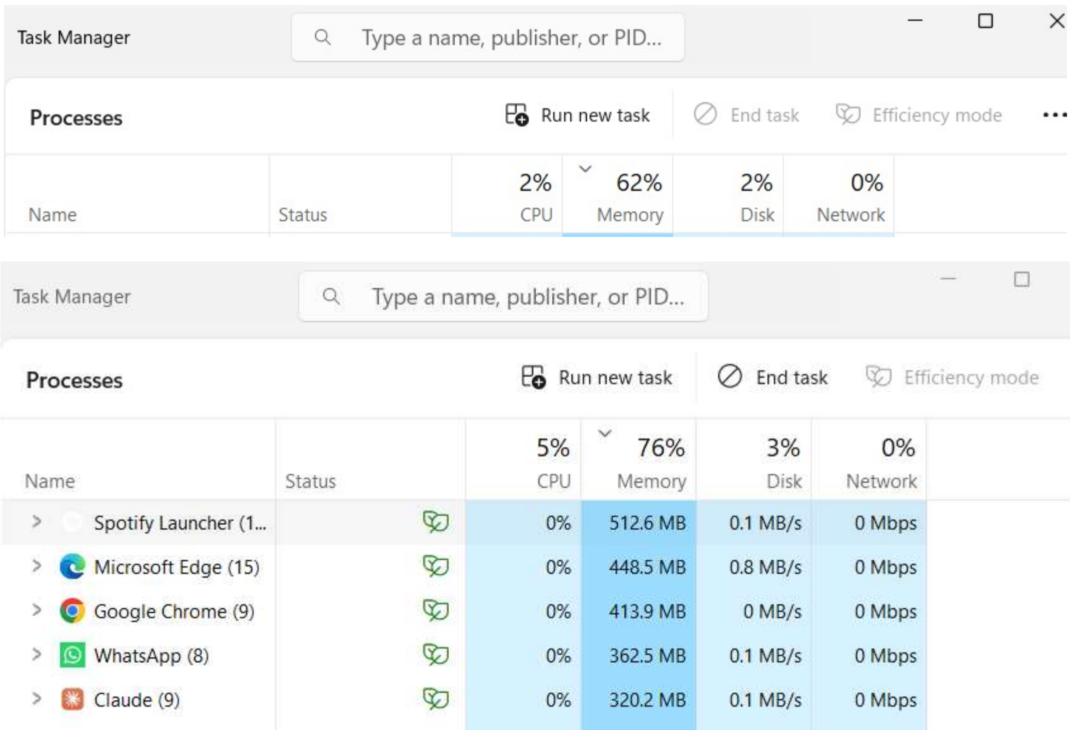
Before opening applications:

- CPU: 2%
- Memory: 62% (approximately 7.4 GB in use)
- Disk: 2%

After opening applications:

- CPU: 5%
- Memory: 76% (approximately 9.1 GB in use)
- Disk: 3%

The most significant change was memory – usage increased by 14 percentage points, adding approximately 1.7 GB of RAM consumption. The newly opened applications consuming the most memory were Spotify (512.6 MB), Microsoft Edge with additional tabs (448.5 MB), and Google Chrome (413.9 MB). CPU usage increased modestly from 2% to 5%, confirming that the bottleneck on this machine is RAM rather than processing power. Disk activity remained low throughout.

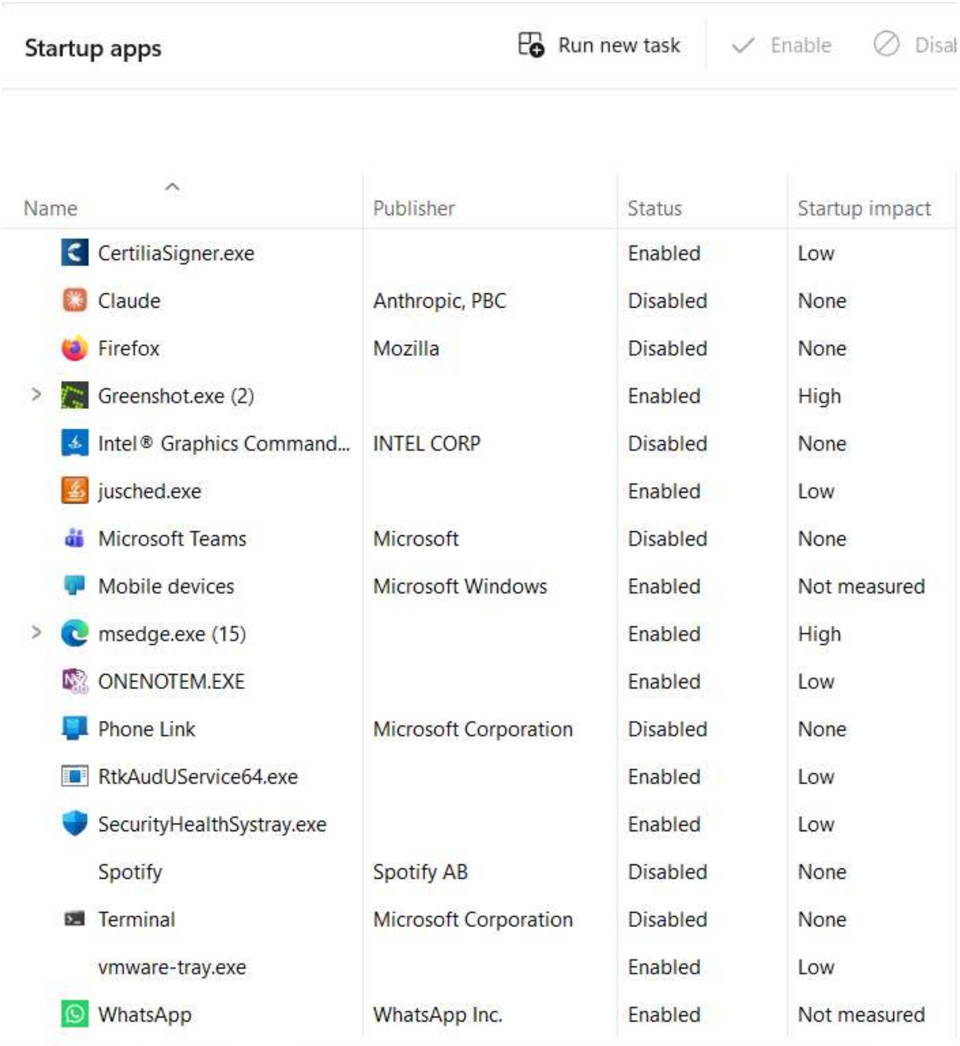


6. **Review the applications configured to start automatically when Windows starts.**

The following applications are configured to start automatically when Windows boots:

Enabled at startup: Greenshot.exe, msedge.exe (15) CertiliaSigner.exe, jusched.exe, ONENOTEM.EXE, RtkAudUService64.exe, SecurityHealthSystray.exe, vmware-tray.exe, Mobile devices, WhatsApp

Disabled at startup: Claude, Firefox, Intel Graphics Command Center, Microsoft Teams, Phone Link, Spotify, Terminal



Name	Publisher	Status	Startup impact
CertiliaSigner.exe		Enabled	Low
Claude	Anthropic, PBC	Disabled	None
Firefox	Mozilla	Disabled	None
Greenshot.exe (2)		Enabled	High
Intel® Graphics Command...	INTEL CORP	Disabled	None
jusched.exe		Enabled	Low
Microsoft Teams	Microsoft	Disabled	None
Mobile devices	Microsoft Windows	Enabled	Not measured
msedge.exe (15)		Enabled	High
ONENOTEM.EXE		Enabled	Low
Phone Link	Microsoft Corporation	Disabled	None
RtkAudUService64.exe		Enabled	Low
SecurityHealthSystray.exe		Enabled	Low
Spotify	Spotify AB	Disabled	None
Terminal	Microsoft Corporation	Disabled	None
vmware-tray.exe		Enabled	Low
WhatsApp	WhatsApp Inc.	Enabled	Not measured

7. **Identify applications that may unnecessarily affect startup performance. Do not disable essential Windows services, antivirus software, or security applications.**

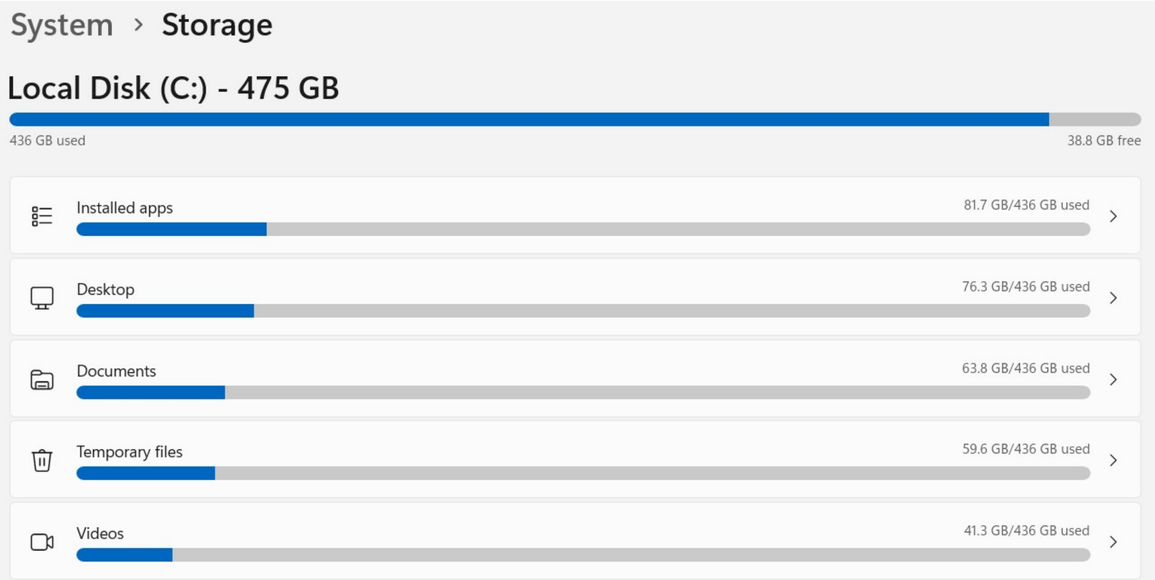
Applications that may unnecessarily affect startup performance:

- Greenshot.exe – screenshot tool marked as High startup impact. Since Greenshot can be launched manually when needed, having it start automatically is unnecessary and adds measurable delay to boot time.

- msedge.exe (15) – Microsoft Edge is marked as High startup impact and launches 15 processes at boot. Unless Edge is used immediately after every startup, this significantly delays boot time with no benefit.
- WhatsApp – marked as Not measured but runs in the background from startup consuming RAM even when not actively used.

8. **Check the available storage space and identify which categories of files or applications are consuming significant disk space.**

The C: drive has a total capacity of 475 GB with 436 GB used and only 38.8 GB free – meaning 91.8% of the drive is occupied. This is a significant concern as Windows performance degrades when free space drops below 10–15% of total capacity, and at the current level there is very little headroom remaining.



Temporary files (59.6 GB) – this is the most actionable category. Temporary files are created by Windows and applications during normal use and are safe to delete.

Desktop (76.3 GB) – storing large files directly on the Desktop

Videos (41.3 GB) – video files are large and ideally should be stored on external drives or cloud storage rather than the system drive.

9. **Based on the information collected, identify at least three possible factors that could affect the performance of your computer.**

1. High RAM usage

With 75–77% of RAM consistently in use even before opening additional applications, the system has very little memory headroom. When applications like Chrome, Spotify, and Edge are opened simultaneously, usage jumps to 76%+ and the system begins relying on the page file. On a machine with only 12 GB RAM and all slots occupied, there is no option to add more memory, making this a fixed constraint.

2. Low available storage space

Only 38.8 GB free out of 475 GB (8.2%) is critically low. Windows needs free space for virtual memory, temporary files, system updates, and normal operations. At this level, the system has almost no room to create temporary files during normal use, which directly slows down application performance and can cause instability.

3. High startup impact applications

Two applications, Greenshot and Microsoft Edge, are marked as High startup impact and launch automatically at boot. Edge alone starts 15 processes at startup. This extends boot time and consumes RAM immediately after startup before the user has even opened anything.

10. **Propose at least five practical actions that could help maintain or improve the computer's performance.**

1. Clean temporary files

59.6 GB of temporary files were identified in the storage analysis. This is the single most impactful action available and would immediately free up space and improve system responsiveness.

2. Disable high-impact startup applications

Disable Greenshot and msedge.exe from the startup list in Task Manager. Both are marked as High startup impact. This will reduce boot time and free up RAM immediately after startup without affecting functionality.

3. Move large files off the C: drive

76.3 GB on the Desktop and 41.3 GB of Videos are stored on the system drive. Moving these to an external drive or cloud storage (OneDrive, Google Drive) would recover over 100 GB and bring free space to a healthy level above 15%.

4. Limit browser tabs and background applications

Chrome (797 MB), Edge (448 MB), and WhatsApp (677 MB) together consume over 1.9 GB of RAM. Using one browser instead of two simultaneously, and keeping tab count low, would meaningfully reduce memory pressure. Browser extensions also consume memory – reviewing and removing unused ones is worthwhile.

5. Run Windows Disk Cleanup or Storage Sense regularly

Enable Storage Sense to automatically delete temporary files, empty the recycle bin, and clean up downloaded files on a schedule. This prevents the current situation from recurring and keeps the drive at a healthy free space level without manual intervention.